

EXECUTIVE SUMMARY

The HealthConnect Clinic Experience Lab project, titled **Improving Patient Appointment Attendance and Healthcare Support Using Data and AI**, focused on using data and AI to understand missed appointments and improve patient support. The Data Analytics work progressed from problem understanding and solution planning in Week 4, through analysis and Power BI dashboard development in Week 5, advanced segmentation and validation in Week 6, testing and refinement in Week 7, and final integration and presentation readiness in Week 8.

Using the same **HealthConnect Appointment Dataset containing 5,000 appointment records and 18 variables from January 2025 to June 2026**, the final analysis identified a **48.46% observed no-show rate**, compared with **46.28% attendance** and **5.26% cancellations**. The analysis found increasing observed no-show rates among appointments with previous no-shows and longer booking lead times. In particular, appointments with **2 or more previous no-shows and booking lead times of 31 days or more** recorded a **72.96% observed no-show rate across 233 appointments**, including 170 no-shows.

The final **Power BI Data Analytics Dashboard** provides validated KPIs, interactive slicers, appointment outcome analysis, previous no-show analysis, booking lead-time analysis, reminder analysis, appointment-type analysis, distance analysis, heatmap visualisations and high-risk segmentation. The dashboard is designed as a **decision-support and monitoring tool**, rather than an automatic patient-level decision system.

The findings provide evidence that can support appointment monitoring, reminder planning, patient-support discussions, investigation of accessibility barriers and future predictive modelling. However, the results represent observed associations and do not establish causation. Important limitations include the fictional and anonymised dataset, some missing values, the unresolved Sunday operating-day inconsistency, interpretation of reminder-channel blanks, and the need for further validation before predictive models are deployed.

The final analytics package also provides inputs for the wider HealthConnect solution by defining validated analytical findings and pre-appointment variables that can support Data Science activities. Although direct cross-track collaboration was limited, the Data Analytics component was independently developed, tested and refined using the available project resources and documented dependencies.

Overall, the Week 8 Data Analytics component is **validated and ready for final integration and presentation**. It delivers a complete Power BI dashboard, validated KPIs, high-risk segmentation, business insights, recommendations, limitations and cross-track inputs. The project demonstrates how structured analytics can transform appointment data into practical evidence for improving healthcare appointment monitoring and patient-support planning.

